Status	Finished
Started	Sunday, 11 October 2026, 4:56 PM
Completed	Sunday, 11 October 2026, 5:00 PM
Duration	4 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It can only be applied to large lists
c. It requires the list to be sorted ✓
d. It does not work with negative numbers

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By sorting the list first
c. By dividing the list into halves
d. By comparing each element in the list sequentially ✓

Question **3**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Linear
b. Binary ✓
c. Hash
d. Both (1) & (3)

Question **4**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and sorted
- b. When the list is small and unsorted ✓
- c. When the list is small and sorted
- d. When the list is large and unsorted

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Binary search
- c. Dynamic search
- d. Hashing

Question **6**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Divide the list into two equal parts
- b. Sort the list
- c. Compare the target element with the first element in the list
- d. Compare the target element with the middle element in the list ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the search algorithm finishes
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **8**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Both (1) & (3)
- c. Binary search
- d. Hash search

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Linear search
- b. Hash search
- c. Binary search ✓
- d. Sequential search

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(1)
- c. O(n log n)
- d. O(log n)

Question **11**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search continues in the right sublist
- b. The search ends successfully ✓
- c. The list is sorted
- d. The search continues in the left sublist

Question **12**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 90 and 99 ✓

c. 90 and 100

d. 94 and 99

Question **13**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It compares elements sequentially

c. It works on unsorted lists

d. It can be applied only if the list is sorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search requires the list to be sorted.

c. Linear search divides the list into halves.

d. Linear search is more efficient than binary search on large lists.

Question **15**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)

b. O(log n)

c. O(1) ✓

d. O🥐